MIL-I-27709A(USAF)
AMENDMENT 4
10 DECEMBER 1986
SUPERSEDING
AMENDMENT 3
1 June 1979

MILITARY SPECIFICATION

INDICATOR, ATTITUDE ARU-14/A, REMOTE

This amendment forms a part of MIL-I-27709A(USAF), dated 13 Sep 1962, and is approved for use within the Department of the Air Force and is available for use by all Department and Agencies of the Department of Defense.

PAGE 1

* Delete "2.1" and add: "2.1 Government documents".

* Add:
"2.1.1 Specifications standards, and handbooks. The following specifications, standard, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation."

* Under SPECIFICATIONS, Federal: delete "QQ-P-416"; add: "PPP-B-601 Boxes, Wood, Cleated Plywood and "PPP-B-636 Box, shipping, fiberboard".

* Under SPECIFICATIONS, Military: Delete: "MIL-E-5272", "MIL-S-6872", MIL-L-6880" and MIL-S-7502". Change "MIL-C-5400" to "MIL-E-5400" and add: "MIL-C-83488 Coating, Aluminum Ion Vapor Deposited".

* Add boxed paragraph at bottom center of page.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Under SPECIFICATIONS, Military: Delete: "MIL-P-7936", "MIL-A-8623", "MIL-A-9067", "MIL-I-26600" and "MIL-D-70327".

* Under STANDARDS, Military, delete "MS33586" and Add:

DOD-STD-100	Engineering Drawing Practices
MIL-STD-454	Standard General Requirements for Electric Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-794	Parts And Equipment, Procedures For Packaging And Packing Of
MIL-STD-810C	Environmental Test Methods"

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MIL-STD-838 Lubrication Of Military Equipment
MIL-STD-889 Dissimilar Metals

* Add 2.2

"2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM D3951 *Packaging, Commercial*

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

PAGE 3

* Delete "3.1" and add:

"3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3)."

3.4.3.1 Line 4 delete "MS33586" and add: "MIL-STD-889."

* Add: "3.4.6 Recovered materials. Recovered materials shall be used to the maximum extent possible without jeopardizing the end use of the item."

* 3.6.g Delete "and dust" and add: "fine dust."

3.6.1, Line 2 delete "MIL-I-26600" and add "MIL-STD-461".

PAGE 11

* 3.12, Line 4 delete "MIL-D-70327" and add: "DOD-STD-100".

3.13.1 Delete and add:

"3.13.1 HERMETIC SEALING - The instrument shall be hermetically sealed. Hermetic sealing is the process by which the indicator is totally enclosed within a suitable metal structure or case and sealed airtight by fusion of metals by welding, brazing, or soldering; fusion of ceramic materials under heat or pressure; or fusion of ceramic materials into a metallic support. Ellastomeric or resinous materials or combinations thereof shall not be used to accomplish hermetic sealing. Gaskets shall not be used to accomplish hermetic sealing."

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Table I: Add title "Roll Servo Connections".

Table II: Add title "Pitch Servo Connections".

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Table III: Under "Connection," opposite pin "A," delete "Neutral (not grounded to case)" and add "Neutral (grounded to case)".

* 3.18 Line 2 and 8 delete "cps" and add: "Hz".

* 3.18.1 Line 3 delete "cps" and add: "Hz."

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3.21 Line 21 delete "MIL-L-6880" and add: "MIL-STD-838."

3.24 Line 2 delete "MIL-S-6872" and add: "MIL-STD-454, requirement 5".

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Delete "3.27.2" and add: "3.27.2 Steel Parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used."

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* Delete "4.1" and add:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements."

"4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material."

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4.2a Delete in its entirety and add: "4.2a First article inspection" (see 4.4)".

4.3.2 Line 2 delete "cps" and add: "Hz."

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4.4 Delete "Qualification" and add: "First Article".

4.4.2. Line 1 delete "qualification test" (two places) and add "First article inspection".

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* 4.5.2.2.d Delete "High altitude-low temperature" and add: "Temperature-altitude."

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* 4.5.2.2.i Delete "Salt spray" and add: "Salt fog."

* 4.5.2.2.j Delete "Sand and dust" and add: "Sand fine dust."

* Under Group II delete "Salt spray", "Sand and dust" and add: "Salt fog" and "Sand fine dust."

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4.6.13.1 Title and line 2: Delete "qualification" and add: "First Article."

* 4.6.15.a Delete "cps" and add: "Hz."

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* 4.6.15.b Delete "cps" and add: "Hz."

* 4.6.15.c Delete "cps" and add: "Hz."

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4.6.20 Line 6 delete "cps" and add: "Hz."

4.6.21 Line 3 delete "MIL-E-5272" and add: "MIL-STD-810C."

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* Delete "4.6.21.4 and add:
"4.6.21.4 HIGH TEMPERATURE EXPOSURE. The indicator shall be subjected to high temperature exposure text in accordance with procedure I of method 501.1 except, perform only steps 1,2, and 3. At the end of step 3 and while still at the high temperature, the indicator shall meet the test specified in 4.6.9."

* Delete "4.6.21.5" and add: "4.6.21.5 HIGH ALTITUDE-LOW TEMPERATURE. The indicator shall be subjected to a temperature-altitude test in accordance with

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Procedure I, equipment category of method 504.1. The time duration for step 1b and step 2 of Table 504.1-II shall be 48 hrs and 24 hrs respectively. Perform only step 16, 2 and 3. The time required for the indicator to start operating shall not exceed 5 minutes. There shall be no binding or sticking at any indication. After the indicator has returned to room temperature and the pressure has increased to atmospheric, it shall be subjected to and shall meet the individual tests."

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* Delete "4.6.21.10" and add: "4.6.21.10 SALT FOG. The salt spray test shall be conducted in accordance with procedure I of method 509.1. There shall be no sticking or binding of the pitch trim knob or damage to the indicator. The tolerances specified under the individual tests shall apply.

* Delete 4.6.21.11" and add: "4.6.21.11 DUST FINE SAND. The sand and dust test shall be conducted in accordance with procedure I of method 510.1. There shall be no sticking or binding of the pitch trim knob."

4.6.21.12 Line 2 delete "procedure V" and add: "procedure I, part 1 category b.1 equipment (curve b) of method 514.2 The range for curve b shall be 5 to 50 Hz and sinusoidal cycling will only be required.

4.6.24 Line 2 delete "MIL-I-26600" and add: "MIL-STD-462."

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Add the following new paragraphs to section 4.

4.7 Inspection of packaging. The inspection of the preservation-packaging and interior package marking shall be in accordance with Group A and B quality conformance inspection requirements, Section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

* Delete Section 5 in its entirety and add:

"5 PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified. (see 6.2)

5.1.1 Level A.

5.1.1.1 Cleaning. Indicators shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Indicators shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

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5.1.1.4. Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method 1C1 of MIL-P-116. Overbox completed pack in PPP-B-636 container. Apply sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Indicators shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified. (see 6.2)

5.2.1 Level A. Units packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Units packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged indicators shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other marking required by the contract or order (see 6.2), Interior and exterior containers shall be marked in accordance with MIL-STD-129."

Delete "6.3" and add: 6.3 When a first article inspection is required sample units shall be inspected as specified (see 3.1 and 4.4).

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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